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Mild Traumatic Brain Injury and Functional Outcome in Older Adults: Pain Interference But Not Cognition Mediates the Relationship Between Traumatic Injury and Functional Difficulties

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Objective: To examine functional status of older people 3 months after mild traumatic brain injury (mTBI) and identify whether pain interference or cognition mediates any relationship found between injury status and functional outcomes. **Setting:** Patients admitted to a Melbourne-based emergency department. **Participants:** Older adults 65 years and older: 40 with mTBI, 66 with orthopedic injury without mTBI (TC), and 47 healthy controls (CC) without injury. **Design:** Observational cohort study. **Main Measures:** Functional outcome was measured using the World Health Organization Disability Assessment Schedule (WHODAS 2.0) and single- and dual-task conditions of the Timed-Up-and-Go task. Pain interference and cognitive performance at 3 months post-injury were examined as mediators of the relationship between injury status (injured vs noninjured) and functional outcome. **Results:** Patients with mTBI and/or orthopedic injury reported greater difficulties in overall functioning, including community participation, compared with noninjured older people (CC group). Both trauma groups walked slower than the CC group on the mobility task, but all groups were similar on the dual-task condition. Pain interference mediated the relationship between injury status and overall functioning [$b = 0.284$; 95% CI = 0.057, 0.536), community participation ($b = 0.259$; 95% CI = 0.051, 0.485), and mobility ($b = 0.116$; 95% CI = 0.019, 0.247). However, cognition did not mediate the relationship between injury status and functional outcomes. **Conclusions:** Three months after mild traumatic injury (with and without mTBI), patients 65 years and older had greater functional difficulties compared with noninjured peers. Pain interference, but not cognition, partially explained the impact of traumatic injury on functional outcomes. This highlights the importance of reducing pain interference for older patients after injury (including mTBI) to support better functional recovery. **Key words:** *aged (MeSH)*, *aging*, *brain concussion (MeSH)*, *brain injuries*, *functional outcome*, *functional status*, *mTBI*, *prognosis*

MILD TRAUMATIC BRAIN INJURY (mTBI) is common among older adults and (as distinct from younger age cohorts) and is often due to falling from standing height.[1] Within the first weeks post-injury, many experience functional impairment.[2,3] Functional status relates to a range of everyday skills (eg, domestic responsibilities and community activities), often categorized into domains such as cognition, self-care, daily living tasks, and sufficient mobility to

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move around independently.[4] These skills are important for independent living[5] and therefore are central in determining outcomes after mTBI in older age, as many will be living alone.

To date, there is limited and mixed evidence about postacute functional outcome specifically in older age cohorts.[3] Some early research suggested that older patients had decreased general functioning compared with younger patients up to 6 months post-injury.[6] Conversely, others indicated generally positive and similar outcome after mTBI for both older and younger adults.[7] Only 67% of older people 65 years and older reported good functional recovery 6 months after mTBI based on Glasgow Outcome Scale scores (GOS-E).[3] The GOS-E is the most used index of functional outcome post-TBI and is useful to document rates of global recovery, especially after moderate-severe injury; however, it has been criticized for lacking sensitivity to detect small but clinically relevant functional change after mTBI.[8] Other measures, such as the widely used World Health Organization Disability Assessment Schedule (WHODAS 2.0), have been developed to capture disability across domains of functioning and have been recommended to identify specific functional difficulties after mTBI.[9,10] Further investigation is needed to better understand the day-to-day experiences of older patients after mTBI using sensitive measures of functional change, and to identify risk factors associated with ongoing difficulties after injury. This can inform clinical management for those at risk of incomplete recovery.

Physical frailty has been identified as a potential risk factor that may impede recovery from mTBI in older age.[11,12] Age-related frailty, more broadly, can be a consequence of multiple health conditions, including chronic pain—with estimates of 60% to 75% of older adults experiencing persistent pain related to chronic health conditions (eg, osteoarthritis).[13] Furthermore, in younger age mTBI cohorts, reporting of pain (eg, from headaches) is not uncommon[14] and persisting or chronic pain can impact functional status.[15] Therefore, further investigation of the significance of pain impacts on functioning in older populations is important. Pain in widely recognized as a complex multidimensional experience and correspondingly has been measured using multiple approaches, often through scaling of pain intensity.[15] An alternative approach, which is relevant to our focus on functional outcome, is to measure "pain interference," which refers to the interference of physical pain on the ability to complete daily activities.[16] Given that presence of pain is frequently reported in noninjured older people,[13] it may be more informative to examine the functional impact of pain after injury compared with noninjured peers to better understand postacute recovery from injury. Additionally, reducing pain interference on daily activities (ie, functional status) may provide a useful target for clinical intervention after injury for those within ongoing symptoms.

Similarly, although recovery from cognitive symptoms is expected 3 months after mTBI in younger age cohorts, older adults are at risk of more persisting cognitive difficulties due to a potential interaction with age-related cognitive changes, particularly slowed processing speed and executive functioning.[17] Preliminary evidence suggests that after mTBI, older adults demonstrate specific difficulties in prospective memory (ie, remembering to do something in the future) compared with noninjured peers.[18] Cognition more generally has also been associated with functional status in a mixed severity sample of older TBI patients.[19] Therefore, the relationship between ongoing cognitive difficulties and functional status in older people following mTBI requires further investigation.

The first aim of this study was to examine functional status of older adults in the nonacute phase following mTBI (ie, 3 months post-injury). We used well-established functional measures expected to be sensitive to change after injury in an older population and compared outcome to older adults with mild traumatic injury but without mTBI, to account for general trauma effects, as well as a noninjured control group. The second aim was to explore whether pain interference and/or cognition mediated the relationship between traumatic injury and functional outcome. It was expected that those with mTBI would report greater functional disability compared with trauma control (TC) patients and noninjured peers, due to the persisting impacts of brain injury. Furthermore, it was expected that pain interference and cognitive status would mediate any relationship found between traumatic injury and functional ability.

METHODS

Participants

This study was approved by the La Trobe University Ethics Committee and Alfred Health Ethics Committee (project ID 382/15). Data were obtained in compliance with institutional/national research standards for human research and the Helsinki Declaration. This study was part of a larger project investigating recovery after mTBI in older adults, some of which has been reported elsewhere.[20]

All participants were 65 years or older and resided within 3 hours of The Alfred Hospital, Melbourne, Australia. Three participant groups were recruited: (i) trauma patients who sustained an mTBI, (ii) trauma patients who sustained a mild traumatic injury (ie, orthopedic injury) but without mTBI, and (iii) noninjured community-dwelling older age adults. General inclusion criteria for all participants included English fluency and functional independence prior to injury, (ie, independence at home). General exclusion criteria for all participants included (i) diagnosed life-threatening medical illness (eg, current cancer diagnosis requiring treatment), (ii) conditions known to affect cognition (eg, Alzheimer's disease), (iii) serious psychiatric illness (eg, chronic schizophrenia), and (iv) hospitalization for a previous significant head injury. All eligible participants underwent a telephone interview, which involved collecting self-reported demographic information (eg, education and medical history) and a cognitive screening questionnaire[21] (exclusion cut-off $<$17).

Trauma patients were recruited at the Alfred Emergency and Trauma Centre between January 2016 and March 2019 after sustaining an mTBI and/or mild traumatic injury. Medical records were screened for participant suitability using general inclusion and exclusion criteria (see previously). A letter outlining the purpose of the study was sent out via post to patients identified as potentially eligible. Patients who did not respond to the letter were followed up via phone call approximately 2 to 3 weeks later. From this, patients either agreed or declined to participate, were ineligible after telephone screening, or were not contactable and did not return our phone call (see flow of recruitment in Supplementary Figure 1, available at: http://links.lww.com/JHTR/A633). Assessment was scheduled for 3 months post-injury for both mTBI and mild trauma comparison (TC) groups.

For mTBI patients, injury was identified using current consensus criteria[22,23] including (i) 1+ symptoms: loss of consciousness of 30 minutes or less, posttraumatic amnesia of 24 hours or less, other transient neurological abnormalities (eg, intracranial lesion not requiring surgery), and/or confusion or disorientation, and (ii) a Glasgow Coma Scale (GCS) score of 13 or more by 30 minutes post-injury or upon later presentation for healthcare. Individuals with abnormal neuroimaging (ie, complicated mTBI) were included as per current mTBI guidelines.[22,24] Inclusion required that symptoms were not due to alcohol, medications, drugs, or caused by treatment for other injuries, other problems (eg, coexisting medical condition), or penetrating craniocerebral injury.[23]

The mild TC group was categorized by an extracranial traumatic injury resulting in (i) a GCS score of 15, (ii) an Abbreviated Injury Scale[25] (AIS) score of 3 or less in any domain, and (iii) no reported confusion surrounding the incident. Patients were excluded if they sustained any AIS score to the head region. AIS scores were used to calculate a total Injury Severity Score[26] (ISS) and peripheral ISS (excluding head region).

Noninjured participants were older adults living independently in the community who volunteered to take part in the study. They were recruited during the same period as trauma patients (January 2016 and March 2019) through researcher networks, advertising, and community talks. Potential volunteers either provided their contact details for further follow-up or contacted researchers using advertised study information.

Measures

Functional outcome

The WHODAS 2.0 measured disability and participation in daily activities over the previous 30 days.[9] The WHODAS 2.0 has good psychometric properties across different cultures and patient populations, including mTBI.[9] As participants were mostly retired from paid employment, we used the 32-item version, which excluded work-related functioning. Participants rated levels of disability for each item on a 5-point Likert scale (none = 0, to extreme = 4). Items were grouped into 6 domains: cognition, mobility, self-care, getting along (sociability), life activities, and community participation. A disability summary score was calculated using the more complex scoring method (see WHODAS 2.0 manual). Scores were converted to standardized z scores, where positive z scores indicated greater functional difficulties (ie, worse outcome).

The Timed-Up-and-Go (TUG) task is a time-based tasking measuring mobility.[27] In the basic condition (TUG_{single}), participants stood up from a chair, walked 3 m (with walking aid if necessary), turned and walk back to the chair, and sat down. The TUG_{single} shows good criterion validity, and good retest and intrarater reliability.[28] The TUG dual-task condition (TUG_{cog}) measured functional dual tasking and required participants to complete the TUG while counting backward from 80 by threes. Completion time was converted to standardized z scores, where positive z scores indicated slower performances. Differences between TUG_{single} and TUG_{cog} conditions were compared by calculating the "dual-task cost"[29] and where positive z scores indicated a greater dual-task cost (ie, greater dual-task difficulties). "Fall risk" was also calculated based on TUG_{single} performance using the recommended cut-off score of more than 13.5 seconds.[27]

Mediators of the relationship between injury status and functional outcomes

Cognition was measured using several neuropsychological tests sensitive to acute symptoms following mTBI[30] (ie, tasks involving processing speed, executive function, and memory) and was described in detail in a previous report.[20] Tests included the WAIS-IV Coding subtest,[31] Trail Making Test,[32] Colour-Word Interference Task,[33] and Hopkins Verbal Learning Test-Revised.[34]

Scores from neuropsychological tests were averaged to calculate a standardized composite z score reflecting overall cognition (NPZ)[4] and where positive z scores indicated better cognitive performance.

Pain interference was measured using a single-item self-report question to determine how much physical pain interfered with daily activities. Single-item questions are shown to be a valid and reliable way to measure perceived health outcomes.[35] Participants rated on a scale (1 = extreme to 5 = not at all) "to what extent do you feel that physical pain prevents you from doing what you need to do?". Ratings were converted to standard z scores where higher scores indicated greater impacts of pain on functioning.

Statistical analysis

Several variables had outliers ($z > 3$) that were skewing the distributions. After transforming these variables, the outliers were all within limits ($z < 3$) and the z residuals for all variables were normally distributed. Variables were transformed using square-root transformation for all WHODAS 2.0 subscales, except "getting along," which used inverse transformation. Square-root transformation was used for TUG dual-task costs, log 10 transformation for TUG_{single}, and inverse transformation for TUG_{cog}. Missing variable analysis was completed using Little MCAR test and did not identify any systematic missing data ($\chi^2 = 9770.867$, $df = 49695$, $P = 1.000$). Therefore, multiple imputation analysis using linear regression was conducted using 5 imputation datasets and then using pooled values from these datasets to replace missing items in the final dataset used for analysis.[36,37] Across the complete dataset, a total of 8% of missing values were imputed.

Two 1-way multivariate analyses of variances (MANOVA) identified group differences (mTBI, TC, and community control [CC]) on WHODAS 2.0 subscales and different TUG conditions. Univariate analyses using Holm alpha adjustments and post hoc testing were conducted for any significant MANOVA. The proportion of those at risk of falls (based on TUG_{single}) for each group was assessed using χ^2 analysis.

Finally, given that both trauma groups performed similarly on functional outcome measures, and to preserve as much statistical power as possible, the 2 trauma groups were combined for the mediation analyses. We used a nominal group as the predictor variable, labeled "injury status" (ie, injured [mTBI and TC] vs noninjured [CC]). After covarying for age, individual mediation analyses examined whether cognition or pain interference mediated the relationship between injury status and outcome measures of interest. Mediation models used a single continuous mediator and continuous outcome variable, and in these situations traditional mediation provides equivalent outcomes as modern mediation methods.[38] All analyses were performed using SPSS Version 27.0 (IBM SPSS Inc, Chicago, Illinois) and a regression PROCESS macro system[39] (version 4.0).

RESULTS

Demographics

One-hundred and forty-two trauma patients agreed to participate. Twenty-one participants were excluded due to injury severity ($n = 10$), alcohol misuse ($n = 2$), neurosurgery after injury ($n = 2$), or incomplete demographic information ($n = 7$). Fifteen participants also withdrew, leaving a total of 106 trauma patients for analysis. Patients were allocated into: (i) mTBI ($n = 40$) and (ii) mild traumatic injury without mTBI (TC; $n = 66$). These were augmented with the group of patients without traumatic injury (CC; $n = 47$).

Differences between injury groups (ie, mTBI, TC, and CC) in relation to gender, age, or level of education were small and nonsignificant (see Table 1). On average, trauma participants were assessed 106 days after injury (range = 71-139 days). The most common mechanism of injury for both trauma groups was falling. Differences in peripheral ISS between trauma groups were small and nonsignificant, but the mTBI group on average did have had a longer hospital stay following injury ($F_{(2,87)} = 25.69$, $P < .001$, $d = -1.123$). Both trauma groups were performing at a lower cognitive level 3 months after injury, as compared with the CC group. Although there was a medium effect between mTBI and CC groups, pain interference was not significantly different between groups after alpha adjustment (see Table 1).

Group differences in functional ability and mobility

A 1-way MANOVA demonstrated a significant main effect for WHODAS 2.0 scores ($F_{(14,290)} = 3.150$, $P \leq .001$, partial $\eta^2 = 0.132$; see Table 2), with analysis of variance (ANOVA) and post hoc testing indicating a medium trauma effect for overall disability ($F_{(2,150)} = 7.110$, $P = .006$, partial $\eta^2 = 0.087$) and large trauma effect for the Participation subscale ($F_{(2,150)} = 15.889$, $P = .001$, partial $\eta^2 = 0.175$). Specifically, trauma patients (mTBI and TC) reported greater difficulties in participation and greater overall disability compared with noninjured older adults. Nonsignificant small to moderate effects were recorded for all other WHODAS 2.0 subscales (see Table 2).

A 1-way MANOVA indicated differences for TUG performances ($F_{(6,298)} = 3.227$, $P = .004$, partial $\eta^2 = 0.061$). Univariate ANOVA demonstrated a medium trauma effect for TUG_{single} ($F_{(2,150)} = 4.421$, $P = .042$, partial $\eta^2 = 0.056$), where the mTBI and TC groups were significantly slower than noninjured older adults (see

TABLE 1 *Information for mild traumatic brain injury (mTBI), trauma comparison (TC), and community control (CC) groups*

	mTBI ($n = 40$)	TC ($n = 66$)	CC ($n = 47$)	P	$d12$	$d23$	$d13$
Demographics							
Age—mean (SD), y	74.70 (6.20)	74.68 (6.74)	73.60 (5.09)	.597	− 0.003	− 0.177	− 0.196
Gender—female, *n* (%)	19 (47.5)	38 (57.6)	27 (57.4)	.549	0.197	0.000	0.200
Years of education—mean (SD)	13.42 (2.60)	13.27 (2.63)	13.36 (2.67)	.957	− 0.057	0.034	− 0.023
Number of comorbidities—mean (SD)	3.58 (2.32)	3.58 (1.97)	3.77 (2.56)	.894	0.000	0.085	0.077
Injury information							
Days since injury—range	108 (87-135)	105 (71-139)	...	.142	− 0.312	...	...
Peripheral ISS—median (IQR)	4 (1-8)	4 (1-8)	...	.582	0.245	...	...
MOI—fall, *n* (%)	29 (72.5)	51 (77.3)	...	.405	0.081	...	...
LOS—mean (SD), d[a]	3.62 (3.32)	0.94 (1.58)	...	<.001[b]	− 1.123	...	...
GCS < 15—yes, *n* (%)	15 (37.5)	0 (0.0)	...	...	...	...	...
LOC—yes, *n* (%)	19 (47.5)	...	...	...	...	...	...
PTA—yes, *n* (%)	16 (40.0)	...	...	...	...	...	...
Abnormal neuroimaging—yes, *n* (%)	20 (50.0)	...	...	...	...	...	...
3-mo outcomes							
NPZ—mean (SD)[c]	− 0.09 (0.66)	− 0.16 (0.65)	0.29 (0.70)	.002	− 0.107	0.670[b]	0.557[b]
Pain interference—mean (SD)[c]	0.30 (0.99)	− 0.01 (0.97)	− 0.24 (0.99)	.044	− 0.317	− 0.235	− 0.545

Abbreviations: d^{12}, Cohen's *d* effect for mTBI versus TC groups; d^{13}, Cohen's *d* effect for mTBI versus CC groups; d^{23}, Cohen's d effect for TC versus CC groups; GCS, Glasgow Coma Scale; IQR, interquartile range; ISS, Injury Severity Score; LOC, loss of consciousness; LOS, length of stay; MOI, mechanism of injury; NPZ, cognitive composite *z* score; PTA, posttraumatic amnesia.

[a]Sample is slightly smaller due to some missing data.

[b]Significant adjusted *P* value < .05; for demographic information Holm alpha adjustment method was used starting at $\alpha/3$; for 3-month outcome Holm alpha adjustment was used starting at $\alpha/2$.

[c]Result previously reported in Hume et al.[20]

TABLE 2 *Z-score mean, SD, and 95% CI for mild traumatic brain injury (mTBI), trauma control (TC), and community control (CC) groups on the subscales and summary score of the WHODAS 2.0*[a]

	Mean (SD) [95% CI]							
	mTBI (*n* = 40)	**TC (*n* = 66)**	**CC (*n* = 47)**	***P***	***d*****12**	***d*****23**	***d*****13**	**Findings**
Disability	0.31 (0.96) [0.01, 0.61]	0.12 (0.98) [−0.12, 0.35]	−0.43 (0.93) [−0.70, 0.15]	.006	0.195	0.573[b]	0.784[b]	mTBI + TC > CC
Cognition	0.25 (0.89) [−0.06, 0.56]	−0.07 (1.03) [−0.31, 0.17]	−0.11 (1.02) [−0.40, 0.17]	.172	0.327	0.039	0.374	Nil
Mobility	0.30 (0.98) [−0.01, 0.60]	(1.00) [−0.23, 0.25]	−0.27 (0.96) [−0.55, 0.02]	.145	0.292	0.285	0.588	Nil
Self-care	0.24 (1.13) [−0.06, 0.55]	0.04 (1.03) [−0.20, 0.28]	−0.27 (0.77) [−0.55, 0.02]	.212	0.187	0.333	0.536	Nil
Getting along	0.16 (0.93) [−0.15, 0.46]	0.11 (1.04) [−0.13, 0.35]	−0.28 (0.96) [−0.57, 0.01]	.198	0.050	0.387	0.465	Nil
Life activities	0.17 (0.96) [−0.14, 0.48]	0.07 (1.09) [−0.18, 0.31]	−0.24 (0.87) [−0.52, 0.05]	.266	0.096	0.309	0.449	Nil
Participation	0.39 (0.99) [0.10, 0.67]	0.20 (0.91) [−0.02, 0.43]	−0.62 (0.85) [−0.88, −0.35]	.001	0.202	*0.918*[b]	*1.102*[b]	mTBI + TC > **CC**

Abbreviations; CI, confidence interval; d^{12}, Cohen's *d* effect for mTBI versus TC groups; d^{13}, Cohen's *d* effect for mTBI versus CC groups; d^{23}, Cohen's *d* effect for TC versus CC groups; SD, standard deviation.

[a]Positive *z* scores indicate greater disability (ie, poorer outcome) on all measures; "Finding" column indicates the differences between all 3 comparison groups (ie, mTBI + TC > CC indicates both mTBI and TC mean *z* scores were significantly lower than the CC group); *P* values are adjusted using Holm adjustment method.

[b]Adjusted $P < .05$; Holm alpha adjustment was used starting at $\alpha/7$.

TABLE 3 *Z-score means and standard deviations for mild traumatic brain injury (mTBI), trauma control (TC), and community control (CC) groups on the TUG_{single}, TUG_{cog}, and DTC*[a]

	Mean (SD) [95% CI]						
	mTBI ($n = 40$)	**TC ($n = 66$)**	**CC ($n = 47$)**	***P***	***d^{12}***	***d^{23}***	***d^{13}***
TUG_{single}	0.27 (1.11) [−0.04, 0.58]	0.07 (1.10) [−0.17, 0.31]	−0.33 (0.65) [−0.61, −0.05]	.042	− 0.181	*− 0.673*[b]	*− 0.426*[b]
TUG_{cog}	−0.05 (0.92) [−0.37, 0.26]	0.14 (1.15) [−0.10, 0.39]	−0.16 (0.81) [−0.45, 0.13]	.268	0.331	− 0.293	− 0.128
DTC	−0.34 (0.90) [−0.65, 0.04]	0.18 (1.07) [−0.06, 0.42]	0.04 (0.92) [−0.25, 0.32]	.058	− 0.515	0.139	− 0.417

Abbreviations: d^{12}, Cohen's *d* effect for mTBI versus TC groups; d^{13}, Cohen's *d* effect for mTBI versus CC groups; d^{23}, Cohen's d effect for TC versus CC groups; DTC, dual-task costs; TUG_{cog}, Timed-Up-and-Go dual task; TUG_{single}, Timed-Up-and-Go single task.
[a] *P* values are adjusted using Holm adjustment.
[b] Adjusted $P < .05$; Holm alpha adjustment was used starting at $\alpha/3$.

Table 3). TUG_{cog} speed was not significantly different between groups ($F_{(2,150)} = 1.329$, $P = .268$, partial $\eta^2 = 0.017$). Group differences in dual-task costs were also not significant ($F_{(2,150)} = 3.662$, $P = .058$, partial $\eta^2 = 0.046$), although there was a medium effect between the mTBI and orthopedic groups.

Mediators of injury status and functional outcome

Given that primary analyses did not identify any significant differences between the 2 trauma groups on any measure, we used a nominal group variable labeled "injury status" (ie, injured [mTBI and TC] vs noninjured [CC]). We examined whether pain interference or cognition mediated the relationship between injury status and (i) overall functional disability, (ii) community participation, and (iii) TUG_{single} performance. Mediation analyses with $N = 5000$ resamples revealed that, after covarying for age, pain interference mediated the relationship between injury status and overall functional disability ($b = 0.284$; 95% CI = 0.057, 0.536), community participation ($b = 0.259$; 95% CI = 0.051, 0.485), and TUG_{single} performance ($b = 0.116$; 95% CI = 0.019, 0.247; see Figure 1), demonstrating that pain interference explained a significant proportion of the relationship between injury and functional outcomes.

After covarying for age, cognition did not mediate the relationship between injury status and overall functional disability ($b = 0.061$; 95% CI = −0.022, 0.167), community participation ($b = 0.039$; 95% CI = −0.045, 0.140), or TUG_{single} performance ($b = 0.065$; 95% CI = −0.015, 0.170).

Although an a priori power analysis suggested it was only possible to enter one covariate in the analysis, a post hoc decision was made to assess whether the mediator remained significant after additionally controlling for gender. Gender as an additional covariate did not alter the results of the mediation analyses already presented.

Group differences in fall risk

Using TUG_{single} performances and a clinical cut-off to identify risk of future falls,[27] χ^2 analysis indicated a significant group difference ($\chi^2_{(2,\ n\ =\ 153)} = 6.169$, $P = .013$, $\Phi = 0.201$). A higher percentage of people in both the trauma groups (mTBI and TC) were at risk of a future fall compared with noninjured older people (see Table 4).

DISCUSSION

This study examined functional status in older people 3 months post-mTBI, compared with patients with mild traumatic injury (without mTBI) and noninjured older people. Injured patients (with and without mTBI) endorsed greater overall disability, especially in community participation, compared with noninjured peers, whereas differences between mTBI and orthopedic trauma patients were not discernible. Both trauma groups demonstrated slower mobility than noninjured older people, but traumatic injury did not adversely impact mobility under dual-task conditions. Pain interference significantly mediated the relationship between injury and functional outcome, whereas cognition did not.

Three months after mild traumatic injury, patients with or without mTBI reported greater overall functional disability compared with noninjured peers. The association between mild injury (with or without mTBI) and functional disability has been previously reported for older people in physical quality of life.[18] These findings suggest that brain injury per se may be less

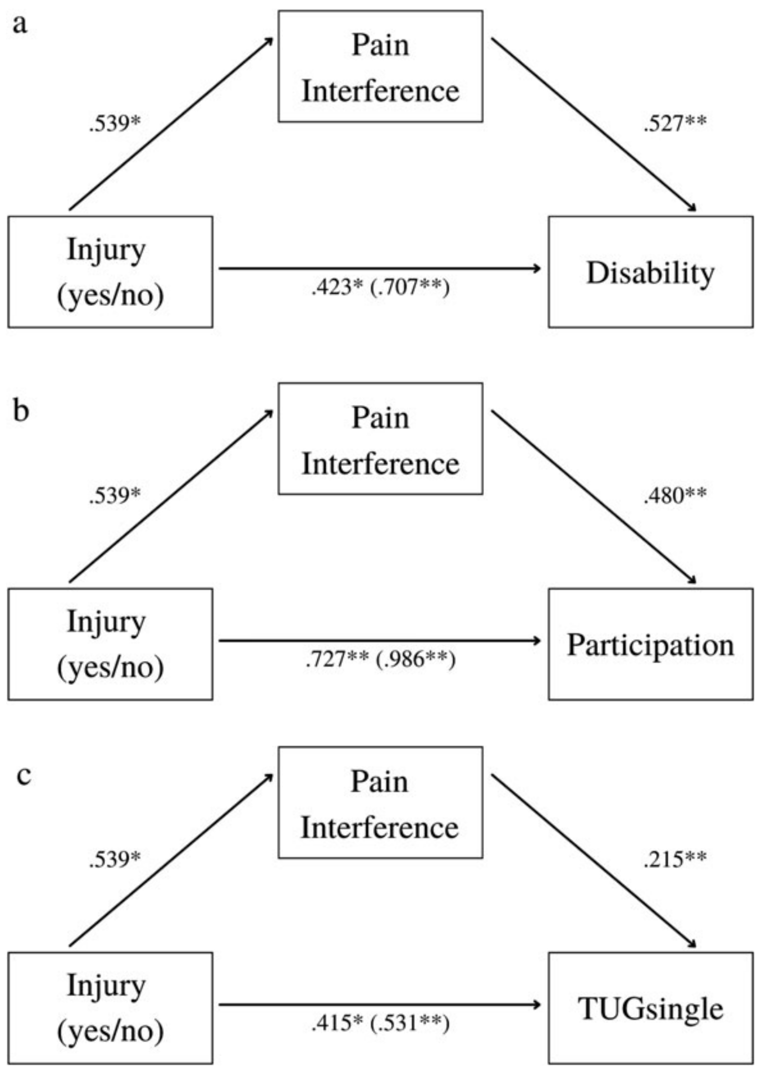

Figure 1. After covarying for age, pain interference mediated the relationship between: (a) presence of injury (yes/no) and overall functional status, (b) presence of injury (yes/no) and community participation, and (c) presence of injury (yes/no) and TUG_{single} performance. *Note:* Numbers refer to the standardized coefficients. Numbers in parentheses depict the beta value when pain scores are not entered as a covariate. $^*P < .05$; $^{**}P < .01$.

important than the experience of traumatic injury more generally in determining functional status after injury. However, in younger age cohorts, there is some evidence of mTBI effects in ongoing global functional impairment 12 months post-injury compared with patients with orthopedic injury,[40] which may indicate delayed longer-term recovery after mTBI specifically. Therefore, further follow-up of functional outcome 6 and 12 months after mTBI is warranted to fully determine the brain injury versus trauma effects in older patients.

The largest difference between trauma groups and noninjured peers was in community participation. This related to perceived barriers and limitations to participation based on the individual's surroundings[9]; however, restrictions in community participation may be a necessary part of injury management and medical care. For example, restricting driving after injury may in turn impact a person's ability to participate in usual community activities such as attending social events and medical

TABLE 4 *Number (%) of people with mild traumatic brain injury (mTBI), traumatic injury (TC), and no injury (CC) at risk of falling based on TUG_{single} performance less than 13.5 seconds*

	Not at risk	At risk	Total
mTBI	30 (75.0)	10 (25.0)	40
TC	55 (83.3)	11 (16.7)	66
CC	45 (95.7)	2 (4.3)	47
Total	130 (85.0)	23 (15.0)	153

appointments. For older people particularly, limitations in community participation enforced by others may be perceived as a removal of independence or evoke feelings of being a burden to relatives. In these cases, psychoeducation about expected timelines for recovery may be important to consider, especially to reduce the possibility of social isolation and reduced quality of life.

After accounting for age, we identified pain interference as a mediator of the relationship between mild traumatic injury (regardless of brain injury) and functional outcome. Of interest, chronic or long-term pain can be a complication after mTBI and mild orthopedic injury, regardless of injury severity.[41,42] Although the mechanisms by which chronic pain develops after mTBI remain unclear,[14] the impact of preexisting pain in older cohorts, such as arthritic pain,[13] combined with even mild traumatic injury (with or without mTBI) likely has a cumulative negative impact on outcome, leading to slower recovery and greater pain interference on daily activities. Pain chronicity in older people could also lead to greater self-restriction in everyday activities due to ongoing discomfort and fear of further injury, similar to reports in younger mTBI samples.[43] For clinicians, the findings highlight the importance of considering how much pain interferes with daily activities after mild injury in older patients. Although some medical services may already prioritize pain management in the early weeks after orthopedic injury, this should be closely monitored in older patients for at least 3 months after injury and include those with mTBI to promote better overall functioning after injury.

As expected, differences in mobility were observed between both trauma groups and noninjured peers. Previous research consistently indicates mobility is impacted after mTBI[44]; however, older adults with orthopedic injury had similar mobility issues, indicating that the additional impact of mTBI was minimal in this sample. Again, the impact of pain interference on functioning mediated the relationship between traumatic injury and mobility. Mobility has also been used to identify older adults at risk of falling[45,46] and in our sample a greater percentage of trauma patients were at risk of future falls ($TUG_{single} > 13.5$ seconds), although it should be noted that various cut-off scores have been recommended.[47] Given that the current study identified pain interference as a mediator of the association between injury and mobility, it raises the possibility that future fall risk could also be related to the impact of pain on mobility. For example, some evidence suggests that older people who report pain are more likely to have fallen in the previous 12 months and are 1.56 times more likely to fall again in future.[48] Therefore, this suggests the importance of effective interventions for pain management for older patients following injury to reduce the risk of future falls.

Cognition did not influence mobility or mediate the relationship between injury and mobility, after accounting for age. This was unexpected given TUG_{single} performance has previously been associated with cognitive skills in various populations.[49,50] Our results suggest that once age-related changes in cognition are controlled for, the influence of cognition on mobility is minimal. Similarly, there were no group differences on cognitive dual-task performance and dual-task costs. This was contrary to expectations. However, a recent study reported that although there were no differences in dual-task costs between younger patients with chronic mTBI symptoms and controls, they did exhibit more variable gait, particularly under the dual-task condition.[51] A further study argued that it is poorer quality of mobility (eg, difficulty turning and greater gait variability) that differentiates younger mTBI patients with persistent balance complaints from noninjured peers.[52] Therefore, quality of gait and balance are important to consider after injury and may be a more accurate measure than simply velocity in the functional mobility of older people in the nonacute phase of injury.

Limitations

We included comparison groups (TC and CC) that were matched for age, education, and gender to control for preexisting demographic differences. All groups were comprehensively screened for relevant preexisting health issues (eg, cognitive impairment and dementia), and all trauma participants were functionally independent prior to injury. Therefore, our sample prior to injury was likely to be a "healthy" cohort of older people. Many older patients who present at emergency departments have underlying comorbidities that will likely impact recovery and mobility after injury.[53] Therefore, the functional difficulties detailed in our study might be an underrepresentation of potential trauma effects in older age. Examining functional recovery in large representative samples would be useful to further inform clinical management after injury. A further issue is that the current study used a single-item measure of pain that specifically examined pain interference on daily activities, whereas cognition was measured using objective neuropsychological testing and composite scores. Although evidence suggests that single-item rating scales are a valid and reliable way to measure perceived health outcomes,[35] future research should consider including validated pain scales that examine other aspects of pain after mild traumatic injury (eg, pain intensity and psychological impacts of pain) as well as interactions with pain medications.[15] Future investigation could also examine the influence and interactions of sleep disturbance and/or frailty on pain and functioning, as these factors have been identified as potentially important for functional outcome.[54] Finally, to extend understanding of the relationship between cognition and outcome in older age, it would be useful to examine self-reported cognitive functioning, as well as cognitive performance after injury, as this may also provide a good target for psychoeducation and rehabilitation after mTBI.

CONCLUSION

Three months after mTBI, adults 65 years and older reported greater functional difficulties and moved more slowly than noninjured peers. However, differences between patients with mTBI and those with mild traumatic injury alone were small and not statistically significant, indicating that these trauma patients share a similar experience after injury. Although both trauma groups performed at a lower cognitive level than the noninjured group, contrary to expectations, cognition had minimal impact on functional status and mobility. Importantly, after accounting for age, pain interference on daily activities partially mediated the relationship between traumatic injury and functional outcomes. These findings highlight the importance of effective management of pain interference on daily activities for older patients after mild traumatic injury (including mTBI), as this has implications for functional outcome at 3-months after injury, and potentially reducing the risk of further falls for these patients.

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